Supplementary Information

Quantifying the Effectiveness of Independent Manual-Valve Verification in Preventing Product-Transfer Risks: A Monte Carlo Process-Safety Assessment

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S1. First suite: all outputs for the 14-valve route

Output	Existing checks	Turn-count collar	Direct sensing	Direct + NFC
Undetected error per transfer	4.027% (3.957–4.098%)	0.454% (0.431–0.479%)	0.302% (0.283–0.322%)	0.285% (0.267–0.305%)
from a valve position	3.206%	0.414%	0.264%	0.247%
from an out-of-sequence step	0.346%	0.026%	0.026%	0.026%
from a post-validation movement	0.503%	0.015%	0.013%	0.013%
Validation with at least one nuisance alarm	0.0%	54.1%	18.6%	17.5%
Validation blocked (device unavailable)	0.0%	8.0%	8.0%	8.0%
Bypass of a blocked validation	0.00%	1.61%	1.61%	1.61%
Nuisance alarms per valve	0.00%	5.37%	1.42%	1.33%
Defects missed by the sensors	–	8.5%	5.5%	5.6%
Detection of a post-validation movement within 60 s	–	96.6%	97.0%	97.0%
Wrong-valve events per 1000 transfers	20.6	20.6	20.6	3.0

Table S1. Outputs of the first suite for the 14-valve route (300,000 transfers per cell; Wilson 95% interval for the primary output).

Tolerance	Direct: alarms	Direct: missed	Turns: alarms	Turns: missed	Direct aged: alarms	Direct aged: missed	Turns aged: alarms	Turns aged: missed
3%	26.7%	1.7%	43.4%	2.1%	28.3%	1.9%	61.7%	4.0%
5%	7.9%	2.0%	20.4%	3.4%	9.6%	2.3%	43.1%	6.3%
6%	4.3%	2.4%	13.2%	4.7%	5.9%	2.8%	36.0%	7.9%
8%	1.4%	5.6%	5.4%	8.3%	2.9%	5.8%	25.9%	11.8%
10%	0.5%	12.2%	2.4%	14.1%	1.9%	13.0%	19.9%	16.6%
12%	0.3%	22.2%	1.4%	22.0%	1.5%	22.4%	16.3%	22.5%
15%	0.2%	35.9%	0.9%	33.7%	1.2%	35.5%	12.8%	31.2%

Table S2. Alarm tolerance sweep (nuisance alarms per valve, share of true defects missed); aged = 60 operations since calibration.

Operations	Direct: alarms per valve	Direct: missed	Turns: alarms per valve	Turns: missed
1	1.4%	5.4%	4.5%	8.5%
3	1.4%	5.6%	5.4%	8.3%
10	1.6%	5.6%	8.2%	9.3%
30	2.1%	5.9%	15.8%	10.7%
100	3.9%	6.0%	36.9%	13.6%
300	8.7%	7.0%	64.7%	16.2%

Table S3. Operations since calibration (tolerance 8%).

Communication loss	Validation blocked	Residual risk, 20% bypass	Residual risk, no bypass
0.0005	4.8%	0.245%	0.208%
0.0010	5.4%	0.250%	0.208%
0.0030	8.0%	0.274%	0.208%
0.0100	16.8%	0.343%	0.208%
0.0200	27.8%	0.454%	0.208%
0.0500	53.4%	0.709%	0.208%

Table S4. Per-device communication-loss probability: validation blocked, and residual risk of the full layer with 20% and 0% bypass.

Complacency	Residual risk
0%	0.229%
10%	0.253%
20%	0.274%
40%	0.325%
60%	0.378%
80%	0.431%

Table S5. Operator complacency (relative reduction of the effectiveness of existing checks) and residual risk of the full layer.

Tap compliance	Wrong-valve events per 1000
0%	20.4
25%	15.8
50%	11.0
75%	6.2
90%	3.1
100%	1.1

Table S6. NFC tap compliance and wrong-valve events per 1000 transfers.

Parameter	Half	Double
existing-barrier miss rate	0.18%	0.46%
skip rate	0.27%	0.28%
incomplete-movement rate	0.25%	0.32%
out-of-sequence rate	0.26%	0.29%
unauthorized-movement rate	0.27%	0.28%
sensor noise	0.23%	0.39%
calibration bias	0.18%	0.48%
alarm tolerance	0.09%	1.96%
device unavailability	0.24%	0.33%
bypass probability	0.24%	0.36%
operator complacency	0.25%	0.33%
wrong-device assignment	0.27%	0.27%

Table S7. One-at-a-time sensitivity of the residual risk of the full layer (base 0.27%) for the parameter at half and double its base value.

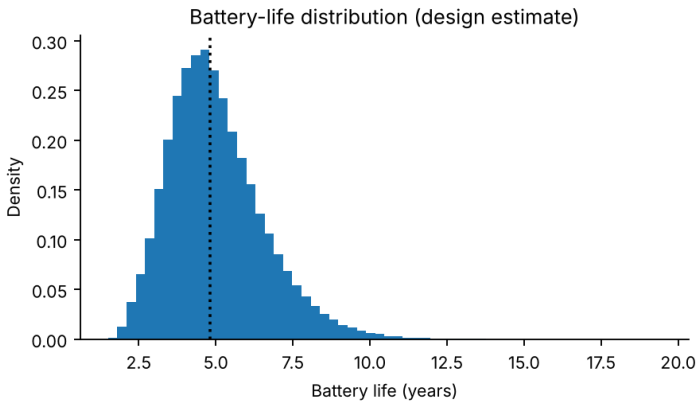


Fig. S1. Design-level battery-life estimate (primary lithium cells, wake-on-motion, 16 s heartbeat): median 4.8 years, 5th to 95th percentile 2.9 to 7.9 years; the analytic mean is 4.7 years.

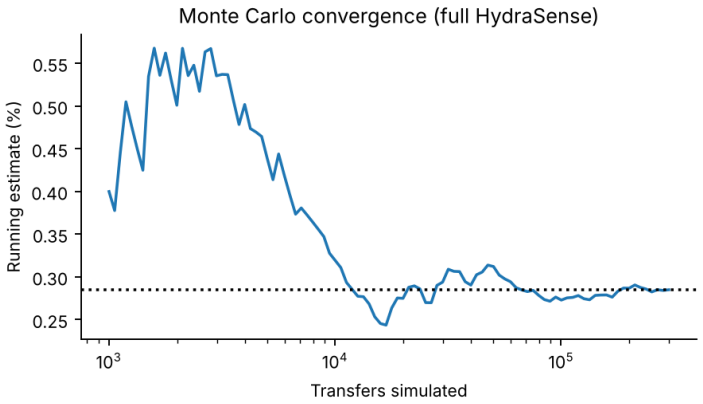


Fig. S2. Convergence of the Monte Carlo estimate of the residual risk with the number of transfers.

S2. Extended studies: all cases

Response	Fatigue coefficient	Residual risk	Risk-reduction factor
100%	0.0	0.275% (0.249–0.302%)	14.8
100%	0.3	0.454% (0.421–0.489%)	8.9
100%	0.6	0.591% (0.553–0.631%)	6.9
90%	0.0	0.720% (0.678–0.764%)	5.6
90%	0.3	0.884% (0.838–0.933%)	4.6
90%	0.6	1.009% (0.960–1.061%)	4.0
80%	0.0	1.145% (1.092–1.200%)	3.5
80%	0.3	1.301% (1.245–1.360%)	3.1
80%	0.6	1.405% (1.347–1.466%)	2.9

Table S8. Operator response to an isolated alarm, fatigue coefficient, residual risk of the full layer (Wilson 95% interval) and risk-reduction factor (existing checks 4.06%).

Response	Fatigue	Tolerance	Residual risk	Nuisance alarms per validation
100%	0.0	4%	0.089%	2.06
100%	0.0	5%	0.102%	1.10
100%	0.0	6%	0.124%	0.59
100%	0.0	7%	0.169%	0.33
100%	0.0	8%	0.275%	0.19
100%	0.0	10%	0.625%	0.06
100%	0.0	12%	1.132%	0.02
100%	0.0	15%	1.791%	0.01
95%	0.3	4%	2.079%	2.06
95%	0.3	5%	1.355%	1.10
95%	0.3	6%	0.898%	0.59
95%	0.3	7%	0.707%	0.33
95%	0.3	8%	0.661%	0.19
95%	0.3	10%	0.878%	0.06
95%	0.3	12%	1.315%	0.02
95%	0.3	15%	1.936%	0.01
95%	0.6	4%	2.939%	2.06
95%	0.6	5%	1.945%	1.10
95%	0.6	6%	1.277%	0.59
95%	0.6	7%	0.929%	0.33
95%	0.6	8%	0.788%	0.19
95%	0.6	10%	0.927%	0.06
95%	0.6	12%	1.329%	0.02

Response	Fatigue	Tolerance	Residual risk	Nuisance alarms per validation
95%	0.6	15%	1.943%	0.01

Table S9. Alarm tolerance with operator response: residual risk of the full layer and nuisance alarms per validation.

Binary tol.	Intermediate tol.	Alarms per binary valve	Alarms per intermediate valve	Alarms per validation	Defects missed	Residual risk
5%	8%	4.67%	8.10%	0.722	5.3%	0.295%
5%	12%	4.67%	0.91%	0.578	20.1%	1.063%
5%	15%	4.67%	0.33%	0.567	30.0%	1.578%
6%	8%	1.81%	8.10%	0.379	5.3%	0.297%
6%	12%	1.81%	0.91%	0.235	20.1%	1.064%
6%	15%	1.81%	0.33%	0.223	30.0%	1.579%
8%	8%	0.31%	8.10%	0.199	5.6%	0.312%
8%	12%	0.31%	0.91%	0.055	20.4%	1.079%
8%	15%	0.31%	0.33%	0.044	30.3%	1.594%

Table S10. Separate tolerances for binary and intermediate valves (direct sensing, 14-valve route).

Operations	Calibration	Nuisance alarms per valve	Defects missed
1	calibration at assignment	1.37%	5.2%
1	re-zero at end stops	1.36%	5.2%
3	calibration at assignment	1.42%	5.6%
3	re-zero at end stops	1.37%	5.6%
10	calibration at assignment	1.60%	5.7%
10	re-zero at end stops	1.39%	5.7%
30	calibration at assignment	2.11%	5.8%
30	re-zero at end stops	1.47%	5.7%
100	calibration at assignment	3.88%	5.8%
100	re-zero at end stops	1.72%	5.6%
300	calibration at assignment	8.69%	7.2%
300	re-zero at end stops	2.35%	6.5%

Table S11. End-stop re-zeroing (direct sensing).

Coverage	Undetected, full layer	Undetected, existing checks	Events, with NFC	Events, without NFC
0%	0.83	5.07	1.38	10.11
25%	0.65	5.07	1.38	10.11
50%	0.45	5.07	1.38	10.11
75%	0.23	5.07	1.38	10.11
100%	0.02	5.07	1.38	10.11

Table S12. Coverage of neighboring valves: undetected collateral events per 1000 transfers (full layer and existing checks), and collateral events (with and without NFC tap-before-touch).

Log-standard deviation	Existing checks (95% interval)	Full layer (95% interval)	Risk-reduction factor
0.0	4.06% (3.99–4.13%)	0.276% (0.258–0.295%)	14.7
0.5	4.04% (3.97–4.11%)	0.276% (0.258–0.295%)	14.7
1.0	4.02% (3.95–4.10%)	0.275% (0.257–0.294%)	14.6
1.5	3.92% (3.85–3.99%)	0.273% (0.255–0.293%)	14.3

Table S13. Common cause: log-standard deviation of the shared multiplier of the operator error rates.

Input	Risk-reduction factor	Full layer risk	Existing-checks risk
existing-barrier miss rate (x)	0.64	0.73	0.96
sensor calibration bias (x)	−0.61	0.61	−0.03
device unavailability (x)	−0.55	0.54	0.05
sensor noise (x)	−0.50	0.54	−0.04
bypass probability	−0.45	0.52	0.05

Input	Risk-reduction factor	Full layer risk	Existing-checks risk
operator error rates (x)	0.42	0.59	0.91
operator complacency	−0.38	0.44	0.03
time-stamp sequence detection	0.28	−0.32	0.06
unauthorized-movement rate (x)	0.25	0.06	0.46
sensor drift (x)	−0.10	0.06	−0.02
out-of-sequence rate (x)	−0.09	0.27	0.32
wrong-device assignment (x)	−0.07	−0.04	−0.08
slip rate (x)	−0.02	−0.04	−0.01

Table S14. Partial rank correlation coefficients of the 13 assumed inputs with the logarithm of the risk-reduction factor, of the residual risk of the full layer and of the residual risk of existing checks.

Readings	Lower	Upper
30	0.796	1.344
50	0.835	1.246
100	0.878	1.162
200	0.911	1.109
500	0.942	1.066

Table S15. 95% interval for the standard deviation of repeated readings, as a multiple of the observed value.

S3. Studies added in version 1.1.0

Sampling uncertainty. 20 independent replications of 150,000 transfers (common random numbers across architectures within a replication) give a residual risk of 4.046% for existing checks and 0.289% for the full layer, a risk-reduction factor of 13.99 (95% bootstrap interval 13.63 to 14.31) and of 13.19 (12.85 to 13.50) for direct sensing without NFC; the relative reduction of residual risk by NFC is 6.1% (5.5% to 6.7%). Slip check: 3,000,000 draws of the sensor error for a defect of 15% of stroke give a miss probability of 0.285% (95% interval 0.279% to 0.292%); the closed form without slip is 0.257% and with slip 0.285%.

Scenario	0.0001	0.0003	0.001	0.003	0.01	0.03	0.1
gateway outage, ignored	14.0	14.0	13.9	13.8	13.5	12.9	10.6
gateway outage, fail-safe blocking	14.0	14.0	14.0	14.0	14.0	14.0	14.0
historian failure, ignored	14.0	14.0	13.9	13.8	13.5	12.9	10.6
historian failure, fail-safe blocking	14.0	14.0	14.0	14.0	14.0	14.0	14.0
wrong valve identification	14.0	14.0	13.9	13.6	13.0	11.3	7.7
calibration procedure error	14.0	14.0	14.0	13.9	13.7	13.4	12.2
all four, outages ignored	13.9	13.9	13.8	13.3	12.0	9.6	5.7
all four, fail-safe blocking	13.9	13.9	13.8	13.6	12.6	10.9	7.3

Table S16. Risk-reduction factor of the full layer against the probability q of a common cause of the sensing layer per validation (existing checks 4.01%; no common cause: 14.0).

Fatigue	Mean response (min)	Time available (min)	Residual risk	Factor	Share from movements
0.0	1	5	0.318%	12.6	10%
0.0	1	10	0.298%	13.4	5%
0.0	1	20	0.293%	13.7	3%
0.0	1	40	0.292%	13.7	3%
0.0	2	5	0.386%	10.4	20%
0.0	2	10	0.340%	11.8	10%
0.0	2	20	0.321%	12.5	4%
0.0	2	40	0.316%	12.7	3%
0.0	3	5	0.546%	7.3	23%
0.0	3	10	0.475%	8.4	11%
0.0	3	20	0.443%	9.0	5%
0.0	3	40	0.434%	9.2	3%

Fatigue	Mean response (min)	Time available (min)	Residual risk	Factor	Share from movements
0.0	5	5	1.010%	4.0	20%
0.0	5	10	0.911%	4.4	11%
0.0	5	20	0.852%	4.7	5%
0.0	5	40	0.830%	4.8	2%
0.0	10	5	2.053%	2.0	15%
0.0	10	10	1.943%	2.1	10%
0.0	10	20	1.845%	2.2	5%
0.0	10	40	1.786%	2.2	2%
0.0	20	5	3.056%	1.3	13%
0.0	20	10	2.971%	1.3	10%
0.0	20	20	2.862%	1.4	7%
0.0	20	40	2.765%	1.4	4%
0.3	1	5	0.321%	12.5	11%
0.3	1	10	0.298%	13.4	5%
0.3	1	20	0.293%	13.7	3%
0.3	1	40	0.292%	13.7	3%
0.3	2	5	0.403%	9.9	21%
0.3	2	10	0.355%	11.3	10%
0.3	2	20	0.333%	12.0	4%
0.3	2	40	0.328%	12.2	3%
0.3	3	5	0.585%	6.9	23%
0.3	3	10	0.509%	7.9	11%
0.3	3	20	0.475%	8.4	5%
0.3	3	40	0.464%	8.6	3%
0.3	5	5	1.072%	3.7	19%
0.3	5	10	0.975%	4.1	11%
0.3	5	20	0.912%	4.4	5%
0.3	5	40	0.887%	4.5	2%
0.3	10	5	2.124%	1.9	15%
0.3	10	10	2.014%	2.0	10%
0.3	10	20	1.918%	2.1	6%
0.3	10	40	1.855%	2.2	2%
0.3	20	5	3.104%	1.3	13%
0.3	20	10	3.022%	1.3	10%
0.3	20	20	2.913%	1.4	7%
0.3	20	40	2.817%	1.4	4%

Table S17. Time-dependent operator response: fatigue coefficient, mean response time (min), median time available after a movement during the transfer (min), residual risk of the full layer, risk-reduction factor and share of the residual risk from movements during the transfer (pre-transfer window 10 min).

Response	Fatigue coefficient	Residual risk	Risk-reduction factor
70%	0.0	1.617%	2.5
70%	0.2	1.698%	2.4
70%	0.4	1.778%	2.2
70%	0.6	1.836%	2.2
70%	0.8	1.881%	2.1
70%	1.0	1.915%	2.1
80%	0.0	1.181%	3.4
80%	0.2	1.287%	3.1
80%	0.4	1.368%	2.9
80%	0.6	1.439%	2.8
80%	0.8	1.486%	2.7
80%	1.0	1.529%	2.6
90%	0.0	0.718%	5.6
90%	0.2	0.844%	4.7
90%	0.4	0.934%	4.3
90%	0.6	1.011%	4.0

Response	Fatigue coefficient	Residual risk	Risk-reduction factor
90%	0.8	1.082%	3.7
90%	1.0	1.120%	3.6
100%	0.0	0.283%	14.1
100%	0.2	0.406%	9.8
100%	0.4	0.518%	7.7
100%	0.6	0.596%	6.7
100%	0.8	0.665%	6.0
100%	1.0	0.715%	5.6

Table S18. Response probability and fatigue coefficient: residual risk of the full layer and risk-reduction factor.

Parameter	Value	Risk-reduction factor
bypass probability	0	18.9
bypass probability	0.1	15.7
bypass probability	0.2	13.6
bypass probability	0.3	12.2
bypass probability	0.5	10.2
bypass probability	0.75	8.3
bypass probability	1	7.0
response probability	0.5	1.6
response probability	0.6	1.9
response probability	0.7	2.5
response probability	0.8	3.5
response probability	0.9	5.4
response probability	0.95	7.9
response probability	1	13.6
common-cause probability (each of four)	0.0001	13.6
common-cause probability (each of four)	0.001	13.4
common-cause probability (each of four)	0.003	12.9
common-cause probability (each of four)	0.01	11.8
common-cause probability (each of four)	0.03	9.2
common-cause probability (each of four)	0.1	5.4
common-cause probability (each of four)	0.3	2.9
communication-loss probability	0.0005	14.8
communication-loss probability	0.003	13.6
communication-loss probability	0.01	10.6
communication-loss probability	0.02	8.0
communication-loss probability	0.05	5.5
communication-loss probability	0.1	4.2
alarm tolerance	0.04	39.0
alarm tolerance	0.06	28.2
alarm tolerance	0.08	13.6
alarm tolerance	0.1	6.3
alarm tolerance	0.12	3.5
alarm tolerance	0.15	2.2

Table S19. Single-parameter sweeps of the risk-reduction factor (14-valve route, other parameters at base values).

Threshold	Tolerance rule	Tolerance	Existing checks	Full layer	Factor	Alarms per validation	Defects missed
5%	fixed 8%	8%	22.32% (22.17–22.47%)	16.089% (15.958–16.221%)	1.4	0.06	59.1%
5%	0.8 x threshold	4%	22.32% (22.17–22.47%)	4.464% (4.391–4.538%)	5.0	1.73	16.0%
10%	fixed 8%	8%	4.07% (4.00–4.14%)	0.292% (0.273–0.312%)	13.9	0.18	6.0%
10%	0.8 x threshold	8%	4.07% (4.00–4.14%)	0.292% (0.273–0.312%)	13.9	0.18	6.0%
15%	fixed 8%	8%	2.53% (2.48–2.59%)	0.059% (0.051–0.069%)	42.7	0.21	1.6%
15%	0.8 x threshold	12%	2.53% (2.48–2.59%)	0.082% (0.072–0.093%)	30.9	0.04	2.1%
20%	fixed 8%	8%	2.20% (2.15–2.26%)	0.054% (0.047–0.063%)	40.6	0.22	1.7%
20%	0.8 x threshold	16%	2.20% (2.15–2.26%)	0.092% (0.081–0.103%)	24.0	0.02	1.8%

Table S20. Defect threshold and alarm tolerance (direct sensing plus NFC).

Design	Intraclass correlation	Design effect	Effective sample size	Acceptance number	Power	Risk at 3%
12 valves x 83 validations	0.00	1.0	996	≤ 20	95.5%	3.5%
12 valves x 83 validations	0.01	1.8	547	≤ 18	82.4%	4.7%
12 valves x 83 validations	0.02	2.6	377	≤ 16	69.8%	4.7%
12 valves x 83 validations	0.05	5.1	195	≤ 12	50.0%	4.9%
12 valves x 83 validations	0.10	9.2	108	≤ 7	34.3%	4.2%
30 valves x 33 validations	0.00	1.0	990	≤ 20	95.7%	3.7%
30 valves x 33 validations	0.01	1.3	750	≤ 19	90.2%	4.1%
30 valves x 33 validations	0.02	1.6	604	≤ 18	83.9%	4.1%
30 valves x 33 validations	0.05	2.6	381	≤ 16	70.2%	4.8%
30 valves x 33 validations	0.10	4.2	236	≤ 13	54.1%	4.8%
71 transfers x 14 valves	0.00	1.0	994	≤ 20	95.6%	3.6%
71 transfers x 14 valves	0.01	1.1	880	≤ 20	94.5%	4.5%
71 transfers x 14 valves	0.02	1.3	789	≤ 19	90.5%	3.6%
71 transfers x 14 valves	0.05	1.6	602	≤ 18	83.5%	4.0%
71 transfers x 14 valves	0.10	2.3	432	≤ 16	70.4%	3.7%

Table S21. Clustered pilot (beta-binomial): design effect, effective sample size, acceptance number, power at 1.4% and risk of accepting 3%.

Coverage	Realistic sensor	Certain detection	Existing checks	Events with NFC	Events without NFC
0%	0.83	0.83	5.07	1.38	10.11
25%	0.65	0.65	5.07	1.38	10.11
50%	0.45	0.45	5.07	1.38	10.11
75%	0.23	0.23	5.07	1.38	10.11
100%	0.02	0.01	5.07	1.38	10.11

Table S22. Coverage of neighboring valves with a realistic sensor (noise, bias, availability) and with certain detection: undetected collateral events per 1000 transfers.

Scenario	P(benign)	P(product)	P(tank)	P(blocked-in)	Existing checks per 1000	Full layer per 1000
required-open valve left closed or partly closed	0.90	0.02	0.00	0.08	19.21	1.34
required-closed valve left open or partly open	0.55	0.30	0.15	0.00	13.63	1.22
out-of-sequence step	0.85	0.05	0.03	0.07	3.70	0.24
movement after validation	0.60	0.20	0.10	0.10	4.79	0.08
collateral movement of a neighbor	0.70	0.15	0.10	0.05	5.25	0.47

Table S23. Event tree (all values assumed or illustrative): conditional probabilities of each consequence given each undetected-error scenario, and scenario probabilities per 1000 transfers (half of the neighbors covered).

Item	Value
benign or recoverable	0
product contamination or misrouting	150,000
tank overflow or release	2,000,000
blocked-in or overpressure	800,000
Transfers per year N_T	1,000

Table S24. Assumed losses per consequence (illustrative currency units) and transfers per year.

S4. Configuration, seeds and tests

Seeds: first suite 2026 (300,000 transfers per case, 120,000 per sweep point); extended studies 2026 (hydrasense_v2.py); version 1.1.0 studies 2027 (hydrasense_v3.py: replications 20270 to 20289; common-cause sweeps seed 21, 300,000 transfers per point; time-response seed 31, 200,000; boundaries seed 41, 200,000; defect threshold seed 51, 300,000; fatigue grid seed 61, 100,000; event tree seed 71, 300,000 per architecture with coverage 0.5 and 4000 draws of the consequence assumptions; second sensitivity analysis 150 draws of 40,000 transfers per case, seeds 7000 to 7149). Base parameters are those of the Params and Arch classes in

HydraSense_MC.py (listed in Table 1 and Section 4); new parameters and defaults are the fields of Params2 in hydrasense_v2.py. Tests: python tests/test_all.py (16 checks; the independent analytic model is tests/independent_check.py).

S5. Reproducibility

The repository (https://github.com/sandlerleon/hydrasense-valve-verification-mc, version 1.1.0) contains the unchanged first-suite model (HydraSense_MC.py), the extended engine and studies (hydrasense_v2.py), the results (results/original_results.json, results/results_v2.json, results/results_v3.json), the figures, the 16 automated tests (tests/test_all.py) and the manuscript sources. Commands: python HydraSense_MC.py --reproduce reproduces the first suite; python hydrasense_v2.py --all runs the extended studies (about 4.5 min on one core), python tests/test_all.py runs the tests; python hydrasense_v3.py --all runs the studies of version 1.1.0 (about 8 min), python make_schematic.py draws Fig. 1. The seed of the extended studies is 2026; every study uses fixed seeds, so rerunning reproduces every number in the text. The code needs Python 3, numpy, scipy and matplotlib.